W13 · THE FOURTH ARABIC NLP CONFERENCE (ArabicNLP 2026) · EMNLP 2026 · BUDAPEST

ARABIC DIALECT SENTIMENT ANALYSIS · DIALECTSENTEVAL 2026 SUBTASK 1

Adversarial Training and Iterative Self-Training

for Zero-Shot Dialect Sentiment Analysis

Momen Ibrahim · Mariam E. Hassib · Ahmed Salem · Marwan Torki

Computer and Systems Engineering Department, Faculty of Engineering, Alexandria University, Egypt | Team ALEXIS

Rank #1 of 12 teams · Macro F1 0.8667 · Accuracy 0.8857

RESULTS AT A GLANCE

First of twelve teams on an unseen dialect

0.8667
Macro F1
official test set, rank 1 of 12

+4.6 pts
Ahead of the runner-up
0.8667 vs 0.8210 Macro F1

0.8857
Accuracy
official test set

52.0%
Lebanese OOV word types
a dialect with zero training rows

The setting

Classify hotel-review sentences as positive, negative or neutral. Training covers four Arabic dialects; the test set adds a fifth, Lebanese, with no labeled data at all.

Our approach

Treat the task as zero-shot transfer: validate leave-one-dialect-out, and choose interventions that match the lexical and orthographic shift.

Three-way sentiment of dialectal hotel reviews

Classify MDS-3 hotel-review sentences as **positive**, **negative** or **neutral**. Training covers four dialects; the released test set adds a fifth, **Lebanese**, with no labeled training data at all.

38.0%
negative

35.2%
positive

26.9%
neutral

Classes are near-balanced and near-uniform across dialects, so dialect gaps reflect transfer difficulty, not label shift.

Dialect	Train	Test
Darija (Moroccan)	433	105
Egyptian	432	105
Jordanian	433	105
Saudi	433	105
Lebanese (unseen)	**0**	**105**
Total	1,731	525

MDS-3 splits per dialect (rows). Lebanese has zero training examples.

THE DIAGNOSIS

A zero-shot transfer problem, not a classification problem

k-fold scores on seen dialects look excellent, but the test set measures something else: transfer to a dialect the model has never seen.

0

training rows for Lebanese

It appears only in the test set: 105 of the 525 test rows.

34.3%

OOV token occurrences

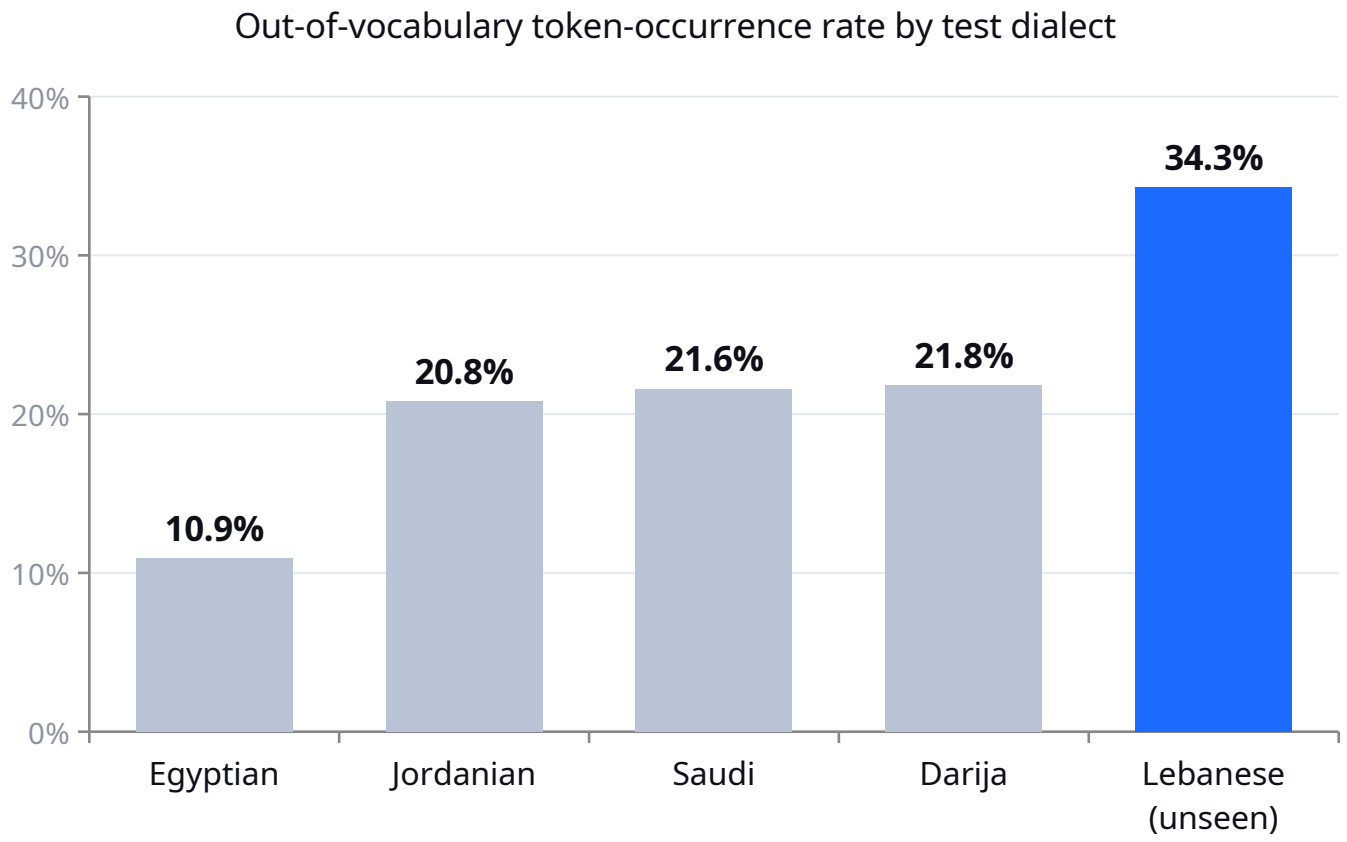
Lebanese, against 10.9–21.8% for the seen dialects; 52.0% of its word types are unseen.

44.4% / 37.8%

negation in negative / neutral

Surface cues separate the classes poorly, so we use contextual encoders, not lexicons or rules.

Lebanese words are mostly unseen in training



Reading the chart

Lebanese reaches 34.3% of token occurrences, more than 1.5× any seen dialect.

Over word types the rate is 52.0%, against 19.5–41.0% for the seen dialects.

A model that memorises dialect-specific function words will fail on Lebanese.

Median length is 16–19 words, so a 128 sub-word cap discards almost nothing.

k-fold cross-validation cannot see an unseen dialect

5-fold cross-validation

StratifiedGroupKFold, grouped by exact text so the 17 duplicated training sentences never straddle folds. Measures generalization to new sentences from seen dialects only.

Leave-one-dialect-out (LODO)

Train on three seen dialects, test on the fourth: simulates the official unseen-dialect condition. Our primary diagnostic; every component is also scored on the leaderboard.

	Dar.	Egy.	Jor.	Sau.	Leb.
LODO 1	**test**	train	train	train	–
LODO 2	train	**test**	train	train	–
LODO 3	train	train	**test**	train	–
LODO 4	train	train	train	**test**	–
Official	train	train	train	train	**test**

Blue: held-out seen dialect in each LODO fold. Amber: the official condition, test on unseen Lebanese.

01

The System

Five components, each aimed at the diagnosed failure

ALEXIS at a glance

1

Diverse encoders

MARBERTv2, CAMeLBERT-DA and AraBERTv2; masked-mean pooling; class-weighted cross-entropy

2

FGM adversarial training

Perturb word embeddings along the loss gradient (ε = 1.0). Largest single gain

3

Iterative self-training

Three rounds on the unlabeled test set; confident committee agreements become pseudo-labels

4

Character noise

Arabic letter-variant swaps plus delete, duplicate and swap at ≈6% per character

5

Uniform ensemble

Softmax average of the three backbones; exact-match lookup for overlapping test rows

Every component was validated leave-one-dialect-out and scored independently on the official leaderboard.

Diverse encoders and adversarial training

MARBERTv2

Strongest single model on the AHASIS pilot of a related dataset

CAMeLBERT-DA

Pretrained on dialectal Arabic; the organizers' sentiment classifier for Subtask 2

AraBERTv2

MSA-centric, included precisely because it errs differently from the other two

Fast Gradient Method (FGM)

At every step a bounded perturbation (ε = 1.0) is added to the word-embedding layer along the direction of the loss gradient, and the model back-propagates through the perturbed pass as well as the clean one.

We read it as a proxy for the robustness a model needs when moving from a seen dialect's embedding neighborhood to an unseen one's.

+0.040 Macro F1 over the gold-only ensemble: the largest single gain

Masked-mean pooling, not [CLS]: a few unfamiliar Lebanese sub-words cannot dominate a single aggregation point.

Self-training and character noise

Iterative self-training (3 rounds)

- A committee of the current best models labels the test set; rows where it agrees above confidence 0.55 join training at 0.7× loss weight.
- Round 3 adds a per-class filter that trims each class's least-confident 10%: a flat threshold under-represents neutral.

386 of 525 rows round 1

499 of 525 rows round 3

Coverage of the test set by pseudo-labels

Character-level noise augmentation



- One noised copy of every gold row: Arabic letter-variant swaps (real orthographic variation) plus delete, duplicate and swap at ≈6% per character.
- Attacks the Lebanese OOV gap directly and makes no assumption about Lebanese class proportions.

Uniform softmax ensemble of the three backbones. The 58 test rows (11.0%) byte-identical to training get an exact-match lookup; models and ablations are judged on the other 89.0%.

02

Results

Official leaderboard, ablation and error analysis

OFFICIAL TEST SET

DialectSentEval 2026 Subtask 1 leaderboard

#	Team / participant	Macro F1
1	**ALEXIS (ours)**	**0.8667**
2	ajwad_hossain	0.8210
3	tasneemduridi	0.8174
4	habiba_ayman	0.8140
5	normalman159	0.8010
6	ahmadalmoustafa	0.7977
7	mona_	0.7953
8	faisalm93	0.7939
9	hoanguitcs	0.7936
10	shadenalzeer	0.7894
11	hh03112005	0.7859
12	23020073	0.7722

Submission IDs omitted.

#1

of 12 teams

first place overall

0.8667

Macro F1

accuracy 0.8857

+4.6

points

ahead of the runner-up (0.8210)

Every step scored on the real leaderboard

Configuration (each row an independent leaderboard submission)	Macro F1	Δ
Baseline: gold-only ensemble	0.7806	–
+ FGM adversarial training (single backbone)	0.8204	+0.040
+ Self-training, 3 rounds	0.8477	+0.027
Forced class-proportion pseudo-label filter (reverted)	**0.8336**	**−0.014**
+ Mild per-class pseudo-label filter	0.8656	+0.018
+ Character-noise augmentation (final system)	**0.8667**	**+0.001**

Δ against the previous retained row; the mild filter is measured from 0.8477 because the forced filter was reverted.

Adversarial training, self-training and the per-class filter each earn their place: +0.040, +0.027 and +0.018 Macro F1.

Failures that confirm the diagnosis

An instructive failure

An aggressive filter forced pseudo-label class proportions to match the four seen dialects. It regressed on the leaderboard (0.8336 vs 0.8477): it assumed Lebanese shares the seen dialects' distribution. The milder, dialect-agnostic filter makes no such assumption, and is what we deployed.

Tried, not deployed

Single-backbone alternative	Score
R-Drop, SWAD, IRM, CORAL, SupCon, MixStyle, char-CNN/BiLSTM	0.785–0.807
Fish (gradient matching)	0.3215
ByT5 backbone (5-fold CV; never submitted)	0.376

None beat FGM's 0.8204, consistent with our diagnosis: they target generic regularization or domain invariance, not the OOV mechanism.

Neutral is the hardest class to recover

Held-out dialect	Neg. F1	Neu. F1	Pos. F1
Jordanian	0.988	0.970	0.983
Saudi	0.976	0.950	0.984
Darija	0.911	0.888	0.933
Egyptian	0.928	0.856	0.914

LODO per-class F1 when each seen dialect is held out.

0.980

Easiest: Jordanian (mean F1)

the Levantine relative of Lebanese

0.899

Hardest: Egyptian (mean F1)

dialect family alone does not fully predict difficulty

An optimistic proxy

LODO rows come from the same translated-review pipeline as training. Seen-dialect CV exceeds 0.93 Macro F1; the official score is 0.8667.

CONCLUSION

What to take away

Diagnose the failure mode first

An unseen test dialect makes this zero-shot transfer: validate leave-one-dialect-out, which k-fold cannot provide.

Match interventions to the shift

FGM and character noise target the lexical and orthographic distance; self-training uses the unlabeled test set.

Rank #1 of 12 teams

Macro F1 0.8667 and accuracy 0.8857, 4.6 points ahead of the runner-up.

OPEN CHALLENGE

- Lebanese remains the dominant error source, even for the best system.
- Seen-dialect CV is above 0.93 Macro F1; the official score is 0.8667.
- Closing the gap will likely require genuine Lebanese-domain signal rather than more regularization.

github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask1-SentimentAnalysis-

Thank You

Questions welcome

Momen Ibrahim · es-momen.ibrahim2019@alexu.edu.eg

Team ALEXIS · Alexandria University, Egypt

PAPER
ACL Anthology

CODE
GitHub

aclanthology.org/2026.arabicnlp-sharedtasks.187
github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask1-SentimentAnalysis-

Key hyperparameters of the deployed system

Hyperparameter	Value
Max sequence length	128 sub-words
Pooling	Masked mean
Loss	Class-weighted cross-entropy
Class weights (neg. / pos. / neu.)	≈ 0.88 / 0.95 / 1.24 (sklearn ‘balanced’, per fold)
FGM perturbation norm ε	1.0
Self-training rounds	3
Committee confidence threshold	0.55
Pseudo-label loss weight	0.7×
Char-noise rate (per character)	≈ 6%
Ensemble combination	Uniform softmax average